CURRICULUM VITAE
SEBASTIANO NICOLUSSI GOLO

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Date of birth:	March 17, 1986
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Me photographed by Tuomas Orponen

Education

Education

Abilitazione Scientifica Nazionale Italiana: 28 February 2025
Settore Concorsuale: 01/A3, Fascia: 2, Quadrimestre n. 3.
Domanda n.126884.

Doctor of Philosophy in Mathematics: 25 October 2017
Double degree based on a cotutelle agreement:
University of Jyväskylä, Finland
Advisor: *Enrico Le Donne*
University of Trento, Italy
Advisor: *Francesco Serra Cassano*
Thesis: *Topics in the geometry of non-Riemannian Lie groups*
Grade: *Pass with distinction*

Master degree in mathematics: March 2012
Università degli Studi di Trento, Italia
Supervisor: *Francesco Serra Cassano*
Thesis: *The tangent space in subRiemannian geometry*
Grade: *110/110 cum laude*

Bachelor degree in mathematics: July 2009
Università degli Studi di Trento, Italia
Thesis: *Symplectic geometry*
Supervisor: *Roberto Pignatelli*
Grade: *108/110*

Employments and Fellowships

Employments and Fellowships

Data Analyst at Oripan srl, Tione (ITA)
March 2026 – Now

Physics Teacher at Liceo Andrea Maffei, Riva del Garda (ITA)
November 2025 – February 2026

Teaching Collaborations (Co.Co.Co.) at the University of Trento (ITA)
September 2025 – August 2026

Senior Researcher (Oberassistent) at the University of Fribourg (CH)
March 2025 – August 2025

Senior Researcher (Senior Forscher) at the University of Fribourg (CH)
May 2024 – February 2025

Postdoctoral Researcher (Tutkijatohtori) at the University of Jyväskylä (FIN)
September 2022 – August 2023
Mentor: Enrico Le Donne

Postdoctoral Researcher (Tutkijatohtori) at the University of Jyväskylä (FIN)
October 2020 – August 2022
Mentor: Katrin Fässler

Postdoctoral Researcher at the University of Padova (ITA)
December 2018 – June 2020
Mentor: Davide Vittone

Postdoctoral Researcher at the University of Birmingham (UK)
December 2017 – November 2018
Mentor: Alessio Martini

Postgraduate Student within the project Marie Curie ITN MAnET (ITA-FIN)
January 2015 – October 2017

Postgraduate Student at the University of Jyväskylä (FIN)
October 2012 – December 2014

Research stays

Research stays

- Theoretical Sciences Visiting Program (TSVP) visitor to OIST, *Okinawa Institute of Science and Technology*, in Japan (August–October 2022)
- Visiting Research Fellow at the Departamento de Geometría y Topología, Universidad de Granada, Spain (November 2017)
- Visiting Graduate Student at the Simons Semester "Symmetry and Geometric Structures", Warsaw, Poland (August–October 2017)
- Visiting Graduate Student at the Department of Mathematics of Tufts University, Medford, USA (January–February 2017)
- Visiting Student at "Laboratoire J.A. Dieudonné", Nice, France (November–December 2016)

Teaching Activities

Teaching Activities

- Physics Teacher at Liceo Andrea Maffei, Riva del Garda (November 2025 – February 2026)
- Tutor for pre-test in mathematics at Economics Faculty of the University of Trento (Fall 2025)
- Teacher of the course "Introduction to PDE"; University of Fribourg, Switzerland (Spring 2025)
- Teacher of the course "Thematical Seminar"; University of Fribourg, Switzerland (Fall 2024)
- Teacher of the Mini-course "BV functions in Carnot groups"; OIST, Japan (August 2022)
- Teacher of the course "Riemannian Geometry"; University of Jyväskylä, (2021–2022)
- Master thesis supervisor; University of Jyväskylä (2021–2022)
- Teaching assistant ("Analysis 1"); University of Padova, Faculty of Engineering (Fall 2019)
- Teaching assistant ("Geometry of surfaces in 3D space"); University of Jyväskylä, Finland (Fall 2012)

Other Activities

Activities

- Co-organizer in the conference "Sub-Riemannian Geometry and Beyond, IV", Schwarzsee, Switzerland (June 2025)
- Co-organizer in the conference "Sub-Riemannian Geometry and Beyond, III", Pisa (June 2023)
- Seal of Excellence from Marie Skłodowska-Curie actions H2020-MSCA-IF-2020: The project proposal *MICA: Minimal Submanifolds in Carnot Groups*, was scored as a high-quality project proposal in a highly competitive evaluation process, with a total score of 92.20%.
- Representative of postdocs in the department council of the Department of Mathematics at the University of Padova (January–June 2020)

- Co-organizer of the workshop "Geometric Measure Theory in Padova", Padova, Italy (January 2020)
- Organizer of seminars "GeA (Geometria e Analisi)" at the Department of Mathematics, University of Padova (Fall 2019)
- Seal of Excellence from Marie Skłodowska-Curie actions H2020-MSCA-IF-2019: The project proposal *MICA: Minimal Submanifolds in Carnot Groups*, was scored as a high-quality project proposal in a highly competitive evaluation process, with a total score of 87.40%.
- Principal investigator (coordinatore) of the GNAMPA–INdAM project "Rectifiability in Carnot Groups" (March 2019 – February 2020)
- Co-organizer of the conference "Harmonic Analysis and PDE Day", Birmingham, UK (September 2018)
- Co-organizer of the conference "Subriemannian Geometry and Beyond", Jyväskylä, Finland (February 2018)
- Organizer of the seminar for doctoral students at Institute Henri Poincaré, Paris (October 2014)
- Organizer of the "Kindergarten Seminar" for doctoral students at the University of Jyväskylä, Finland (Spring 2014)

Talks since 2015

Talks

- "Embeddings of Nilpotent groups into L1", Geometry and analysis of groups and manifolds, Centro De Giorgi, Pisa, Italy (June 2023)
- "Horizontal jet spaces on Carnot groups", Geometry Seminar, University of Fribourg, Switzerland (December 2022)
- "Sub-Riemannian geometry is everywhere!", TVSP seminar, OIST, Okinawa, Japan (September 2022)
- "Uniform measures", Geometry and Differential Equations Seminar, Impan, Warsaw, Poland (June 2022)
- "Uniform measures", Analysis Seminar, University of Jyväskylä, Finland (April 2022)
- "Horizontal jet spaces on Carnot groups", Geometry and Differential Equations Seminar, Impan, Warsaw, Poland (November 2021)
- "Horizontal jet spaces on Carnot groups", 2021 AGENT–Forum, Aalto University, Finland (October 2021)
- "From Peano to Bernstein in one seminar", Geometry Seminar, University of Jyväskylä, Finland (November 2020)
- "Spectral multipliers and wave equation for sub-Laplacians", Latest in Geometric Analysis; Celebration of Pekka Koskela's 59th birthday, Będlewo, Poland (November 2019)
- "Minimal surfaces in the Heisenberg group", IMPAN, Warsaw (October 2019)
- "From metric spaces to Lie groups", IMPAN, Warsaw (October 2019)
- "Tangent spaces", IMPAN, Warsaw (October 2019)
- "Spectral multipliers and wave equation for sub-Laplacians", Analysis Seminar, IMPAN, Warsaw (May 2019)
- "Metric Lie groups and their dilations", Some topics of Geometric Analysis and Geometric Measure Theory, Pisa, Italy (April 2019)
- "Spectral multipliers and wave equation for sub-Laplacians", Subriemannian Geometry and Beyond II, Jyväskylä, Finland (February 2019)
- "Spectral multipliers and wave equation for sub-Laplacians", Analysis Seminar, Padova, Italy (January 2019)
- "Metric Lie groups and their dilations", Geometry Seminar, Bristol, UK (November 2018)
- "Spectral multipliers and wave equation for sub-Laplacians", Analysis Seminar, Birmingham, UK (November 2018)
- "Minimal surfaces in the Heisenberg group", MAnET final meeting, Bologna, Italy (January 2018)
- Talk "Minimal surfaces in the Heisenberg group", University of Birmingham, UK (December 2017)
- Talk "Minimal surfaces in the Heisenberg group", University of Granada, Spain (November 2017)
- Talk "Minimal surfaces in the Heisenberg group", IMPAN, Warsaw, Poland (October 2017)

- Talk "Homogeneous distances and shape of spheres", IMPAN, Warsaw, Poland (October 2017)
- Talk: "Topics in non-Riemannian geometry", Public defence of my Doctoral Thesis, University of Jyväskylä, Finland (September 2017)
- Talk: "Characterization of self-similar Lie groups", Graduate seminar, University of Jyväskylä, Finland (March 2017)
- Talk: "Characterization of self-similar Lie groups", University of Trento, Italy (March 2017)
- Talk: "The PDEs of minimal surfaces in the Heisenberg group", Worcester University, MI, USA (February 2017)
- Colloquium: "From homogeneous metric spaces to Lie groups", Tufts University, Medford, MI, USA (February 2017)
- Talk: "A new stable surface in the Heisenberg group", MAnET Metric Analysis Meeting, Université Paris-Sud, France (January 2017)
- Talk: "Geometry and analysis on Carnot groups", University of Trento, Italy (September 2016)
- Talk: "Self-similar groups and their spheres", Aalto University, Finland (September 2016)
- Talk: "Groups with polynomial growth", University of Jyväskylä, Finland (March 2016)
- Talk: "A stability result for nonminimal surfaces in the Heisenberg group", University of Jyväskylä, Finland (March 2016)
- Talk: "An insight into Carnot-Carathéodory Geometry", at the Joint PhD Opening 2016 Biomolecular Sciences and Mathematics, Trento, Italy (February 2016)
- Talk: "Geometry and analysis in the Heisenberg group: results and further problems", University of Trento, Italy (October 2015)
- Talk: "Regularity of spheres in metric groups", for the Junior seminars at ICMAT, Madrid, Spain (May 2015)
- Talk: "Regularity of spheres in metric groups", University of Trento, Italy (March 2015)
- Talk: "Regularity of spheres in metric groups", at the Geometric Seminar, University of Jyväskylä, Finland (February 2015)

Publications

[32*] E. Le Donne, S. Nicolussi Golo, and N. Paddeu. Normal Curves in Sub-Finsler Lie Groups: Branching for Strongly Convex Norms and Face Stability for Polyhedral Norms. *arXiv e-prints.*

[31*] K. Fässler, E. Le Donne, S. Nicolussi Golo, A. Ottazzi, and P. Pansu. Quasiconformal VS quasi-isometric equivalence in spaces with controlled growth. *arXiv e-prints.*

[30*] S. Nicolussi Golo, F. Serra Cassano, and M. Vedovato. The Bernstein problem for Sobolev intrinsic graphs in the Heisenberg group. *arXiv e-prints.*

[29*] E. Le Donne, L. Nalon, S. Nicolussi Golo, and S. Y. Ryoo. Asymptotics of Riemannian Lie groups with nilpotency step 2. *arXiv e-prints.*

[28*] A. Kijowski, S. Nicolussi Golo, and B. Warhurst. Equivalence of sub-Laplacian on Polarized groups. *arXiv e-prints.*

[27*] S. Nicolussi Golo and B. Warhurst. Moduli of Sub-Laplacians on the second Heisenberg group. *arXiv e-prints.*

[26] C. Y. Guo, S. Nicolussi Golo, M. Williams, and Y. Xuan. Quasiregular mappings between subRiemannian manifolds. *J. Lond. Math. Soc. (2) 112 (2025), no. 2, Paper No. e70254.*

[25] M. Di Marco, A. Julia, S. Nicolussi Golo, and D. Vittone. Submanifolds with boundary and Stokes' Theorem in Heisenberg groups. *Trans. Amer. Math. Soc. 378 (2025), 4955-4990.*

[24] M. G. Cowling, V. Kivioja, E. Le Donne, S. Nicolussi Golo, and A. Ottazzi. From homogeneous metric spaces to Lie groups. *C. R. Math. Acad. Sci. Paris 362 (2024), 943–1014.*

[23] S. Nicolussi Golo and Y. Zhang. Curvature Exponent and Geodesic Dimension on Sard-regular Carnot groups. *Anal. and Geom. in Metr. Spaces, vol. 12, no. 1, pp. 20240004* (2024).

[22] A. Ottazzi, and S. Nicolussi Golo. Left-invariant distributions diffeomorphic to flat distributions. *Geom. Dedicata 218, 56* (2024).

[21] L. Ambrosio, S. Nicolussi Golo, and F. Serra Cassano. Classical flows of vector fields with exponential or sub-exponential summability. *J. Differ. Equations 372 458–504* (2023).

[20] L. Ambrosio, S. Nicolussi Golo, and F. Serra Cassano. Optimal C^∞-approximation of functions with exponentially or sub-exponentially integrable derivative. *Calc. Var. Partial Differ. Equ. 62 21* (2023).

[19] S. Nicolussi Golo, and B. Warhurst. Jet spaces over Carnot groups. *Rev. Mat. Iberoam. 39 (2023), no. 6, pp. 2289–2330.*

[18] S. Eriksson-Bique, C. Gartland, E. Le Donne, L. Naples, and S. Nicolussi Golo. Nilpotent groups and biLipschitz embeddings into L1. *International Mathematics Research Notices, Volume 2023, Issue 12, June 2023, Pages 10759–10797.*

[17] G. Antonelli, E. Le Donne, and S. Nicolussi Golo. Lipschitz Carnot-Carathéodory structures and their limits. *J. Dyn. Control. Syst. 29, 805–854* (2023).

[16] A. Julia, S. Nicolussi Golo, and D. Vittone. Lipschitz functions on submanifolds in Heisenberg groups. *International Mathematics Research Notices, Volume 2023, Issue 9, May 2023, Pages 7399–7422.*

[15] E. Le Donne, V. Kivioja, and S. Nicolussi Golo. Metric equivalences of Heintze groups and applications to classifications in low dimension. *Illinois J. Math.* 66 (1) 91 - 121, April 2022.

[14] A. Julia, S. Nicolussi Golo, and D. Vittone. Area of intrinsic graphs and coarea formula in Carnot groups. *Math. Z.* 301, 1369–1406 (2022).

[13] A. Martini, D. Müller, and S. Nicolussi Golo. Spectral multipliers and wave equation for sub-Laplacians: lower bounds of Euclidean type. *J. Eur. Math. Soc. (JEMS)* (2021).

[12] A. Julia, S. Nicolussi Golo, and D. Vittone. Nowhere differentiable intrinsic Lipschitz graphs. *Bull. Lond. Math. Soc.*, 53(6) (2021).

[11] A. Julia and S. Nicolussi Golo. Intrinsic rectifiability via flat cones in the Heisenberg group. *Ann. Sc. Norm. Super. Pisa Cl. Sci.*, Vol. XXIII, Issue 2 (2022).

[10] N. Fisher and S. Nicolussi Golo. Sub-Finsler horofunction boundaries of the Heisenberg group. *Anal. Geom. Metr. Spaces*, 9(1):19–52, 2021.

[9] S. Nicolussi Golo and M. Ritoré. Area-minimizing cones in the Heisenberg group H. *Ann. Fenn. Math.*, 46(2):945–956, 2021.

[8] E. Le Donne and S. Nicolussi Golo. Metric Lie groups admitting dilations. *Ark. Mat.*, 59(1) (2021).

[7] S. Nicolussi and F.Serra Cassano. The Bernstein problem for Lipschitz intrinsic graphs in the Heisenberg group. *Calc. Var. Partial Differential Equations*, 58(4):Art. 141, 28, 2019.

[6] S. Golo and S. Rigot. The Besicovitch covering property in the Heisenberg group revisited. *J. Geom. Anal.*, 29(4):3345–3383, 2019.

[5] S. Golo. Some remarks on contact variations in the first Heisenberg group. *Ann. Acad. Sci. Fenn. Math.*, 43(1):311–335, 2018.

[4] E. Le Donne and S. Nicolussi Golo. Regularity properties of spheres in homogeneous groups. *Trans. Amer. Math. Soc.*, 370(3):2057–2084, 2018.

[3] S. Nicolussi Golo. Topics in the geometry of non-Riemannian Lie groups. PhD thesis, *Report*, Department of Mathematics and Statistics, University of Jyväskylä, 163 (2017).

[2] E. Le Donne, S. Nicolussi Golo, and A. Sambusetti. Asymptotic behavior of the Riemannian Heisenberg group and its horoboundary. *Ann. Mat. Pura Appl. (4)*, 196(4):1251–1272, 2017.

[1] F. Bigolin and S. Nicolussi Golo. A historical account on characterizations of C^1-manifolds in Euclidean spaces by tangent cones. *J. Math. Anal. Appl.*, 412(1):63–76, 2014.

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